

Deshan Lokuge

NETWORKING | SECURITY ENGINEER

Galle, Sri Lanka

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Summary

Electrical and Information Engineering undergraduate specializing in computer networking, telecommunications and security, with internship experience in telecom network operations and IoT deployment. Experienced in designing and testing enterprise networks, including network monitoring, automation and security. Seeking opportunities in network engineering, network security and NOC operations.

Education

- BSc. Eng (Hons.) Electrical and Information Engineering 2022 – 2026

University of Ruhuna

GPA: 3.64

Focus: Networking, Telecommunications and Security | Minor: Electrical Engineering

Selected Coursework: Telecommunication and Computer Networking; Operating Systems; Information Security; Data Networks and Design; Embedded Systems Design

- G.C.E. Advanced Level Examination

2020

Richmond College, Galle

Experience

IoT & Automation Engineering Intern — Samson International PLC | Repo | **Aug–Oct 2025**

- Designed and deployed a production-monitoring system across press machines, enabling real-time monitoring through ThingsBoard web and mobile dashboards and remote configuration.
- Improved IoT device security by using access tokens and separate Wi-Fi and OTA credentials, while adding automatic MQTT reconnection for reliable communication.
- Designed custom PCBs and enhanced system reliability through sensor input isolation and signal filtering.

Network Operations & IoT Engineering Intern — Sri Lanka Telecom PLC | Repo | **May–Aug 2024**

- Performed network monitoring and maintenance at SLT OPMC, Galle, through fault identification and troubleshooting across operational telecom infrastructure.
- Contributed to the Fazenda Smart Agro project at SLT Digital Lab, collaborating with engineers on planning, testing, technical documentation and system refinement.
- Integrated a microcontroller-based smart irrigation system with ThingsBoard dashboards, calibrated sensors and tested real-time monitoring, alerts and remote control.

Technical Skills

Networking & Telecom: GNS3, Packet Tracer, CML, Wireshark, Zabbix, Open-RAN, srsRAN, Near-RT-RIC

Systems & Cloud: Linux/Ubuntu, Docker, Kubernetes, VMWare

Security: Falco, SPIFFE/SPIRE, RBAC, Calico CNI, Cosign, GPG

Programming & Automation: Python, C/C++, Bash, Ansible, SQL/MySQL, Git/GitHub, React.js

IoT & Embedded: Arduino, MQTT, ThingsBoard, PCB Design, EasyEDA, Proteus, Microchip Studio

Software & Tools: VS Code, draw.io, Jira, MATLAB, Figma, Grafana

Languages

Sinhala: Native

English: Professional working proficiency

Projects

1. **Threat Containment Strategies for Compromised xApps in Open RAN** | Repo
 - Developed ZT-XGuard, a zero-trust runtime workload attestation framework with SPIFFE/SPIRE and deployed a Kubernetes testbed integrating OSC Near-RT RIC, Open5GS, srsRAN and operational xApps.
 - Integrated a secure xApp onboarding pipeline into the Near-RT RIC using GPG, Cosign and short-lived X.509 SVIDs to strengthen CIA-triad security for identity management.
 - Designed a MEWMA, Hotelling-T² resource anomaly detector for DoS attacks, achieving a 98.07% true positive rate. Integrated Falco for kernel-level container inspection with 230 ms detection latency and automated compromised xApp microsegmentation using Calico CNI within 2–4 seconds.
 - Resolved xApp E2 node connectivity in NIST's O-RAN Testbed Automation GitHub repository and documented the fix in GitHub Issue.
2. **Enterprise Campus Network — Design, Automation & Monitoring** | Repo
 - Designed and implemented a 3-tier hierarchical campus network using 10 Cisco IOS devices in GNS3, with OSPF, PAT and ACLs across 4 departmental VLANs.
 - Automated router and switch configurations using Python/Netmiko and Ansible, achieving zero configuration drift and developing rollback playbooks for recovery.
 - Implemented centralized Zabbix monitoring over SNMP for real-time visibility, fault detection and performance analysis.
3. **Offensive and Defensive PEN Tool** | Repo
 - Designed and implemented a portable wireless penetration testing tool for network security assessments. Features included Wi-Fi/Bluetooth scanning, MAC sniffing, Evil Twin and Portal attacks, packet monitoring, and deauthentication attack detection.
 - Applied penetration testing methods and wireless security protocols to perform offensive and defensive security evaluations focused on PCAP analysis and traffic monitoring.
4. **PayFlow - Online Voucher Application Portal** | Repo
 - Developed a web-based application to streamline reimbursement and voucher requests, improving financial management for university academic and non-academic staff.
 - Acted as Scrum Master, managing Jira boards, and contributed to frontend development, including Clerk-based authenticated user logins using React.js.

Certifications and Licenses

- Cisco Networking Academy — CCNA v7 Course: ITN | SRWE | ENSA
- Computer Networks - Huawei
- Endpoint Security - Cisco
- Introduction to Cybersecurity - Cisco
- Java and Python Programming

Volunteering and Co - Curricular Involvement

- Volunteered for the Mehewara project organized by second-year undergraduates of the faculty, aimed at supporting O/L students.
- ReXtro 2025 by preparing and demonstrating the Luminex FTTH project exhibit in Networking & Telecommunication zone during the Silver Jubilee Engineering Exhibition.
- Volunteer member of the ECSL Electrician's Guide team, University of Ruhuna Team.
- Membership - Institution of Engineers, Sri Lanka (IESL) - Membership number: S-32500

References

Dr. Chatura Seneviratne | Head, DEIE, University of Ruhuna | chatura@eie.ruh.ac.lk

Eng. Ilumini Muthunaike | R&D Engineer, Sri Lanka Telecom PLC | ilumini@slt.com.lk